Theorem 6.5

For all real number $r \neq 1$. For any $n \in \mathbb{N}$,

$$1 + r + r^2 + ... + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Proof.

We proceed by induction, so there are two cases for n.

(i). Consider when $n = 1$. Now $1 + r = \frac{1-r^2}{1-r} = 1 + r$. So, our original statement is true for all $r \neq 1$ when $n = 1$.

(ii). Assume there exists some arbitrary $n \in N$ such that our original statement is true. Now, consider

$$(1 + r + r^2 + ... + r^n) + r^{n+1}.$$

By our assumption, we know

$$\frac{1 - r^{n+1}}{1 - r} + r^{n+1} = \frac{1 - r^{n+1}}{1 - r} + \frac{r^{n+1}(1 - r)}{1 - r} = \frac{1 - r^{n+1}}{1 - r} + \frac{r^{n+1} - r^{n+2}}{1 - r} = \frac{1 - r^{n+2}}{1 - r}.$$

Therefore, $1 + r + r^2 + ... + r^{n+1} = \frac{1-r^{n+2}}{1-r}$ is true whenever $1 + r + r^2 + ... + r^n = \frac{1-r^{n+1}}{1-r}$ is true.

Thus, by the PMI, our original statement is true for all $n \in \mathbb{N}$.

■

Theorem 6.7

For all $n \in \mathbb{N}$, then P(n) := "then 3 divides $4^n - 1$".

Proof.

We proceed by induction, so there are two cases for n.

(i). Consider when $n = 1$. Then, $3|3$. We know any number divides itself. Therefore, our original statement is true when $n = 1$.

(ii). Assume there exists some arbitrary $n \in N$ such that 3 divides $4^n - 1$. Now consider $4^{n+1} - 1$. We then know

$$4^{n+1} - 1 = 4(4^n) - 1 = 4^n + 4^n + 4^n + (4^n - 1) = 3(4^n) + (4^n - 1).$$

Observe that **Theorem 2.15** implies 3 divides $3|3(4^n)$. Subsequently, by our assumption, we know 3 divides $4^n - 1$. Thus, **Theorem 2.18** implies $3|(4^{n+1} - 1)$. Therefore, 3 divides $4^{n+1} - 1$ whenever 3 divides $4^n - 1$.

Now, by the PMI, for all $n \in \mathbb{N}$, then 3 divides $4^n - 1$.

■